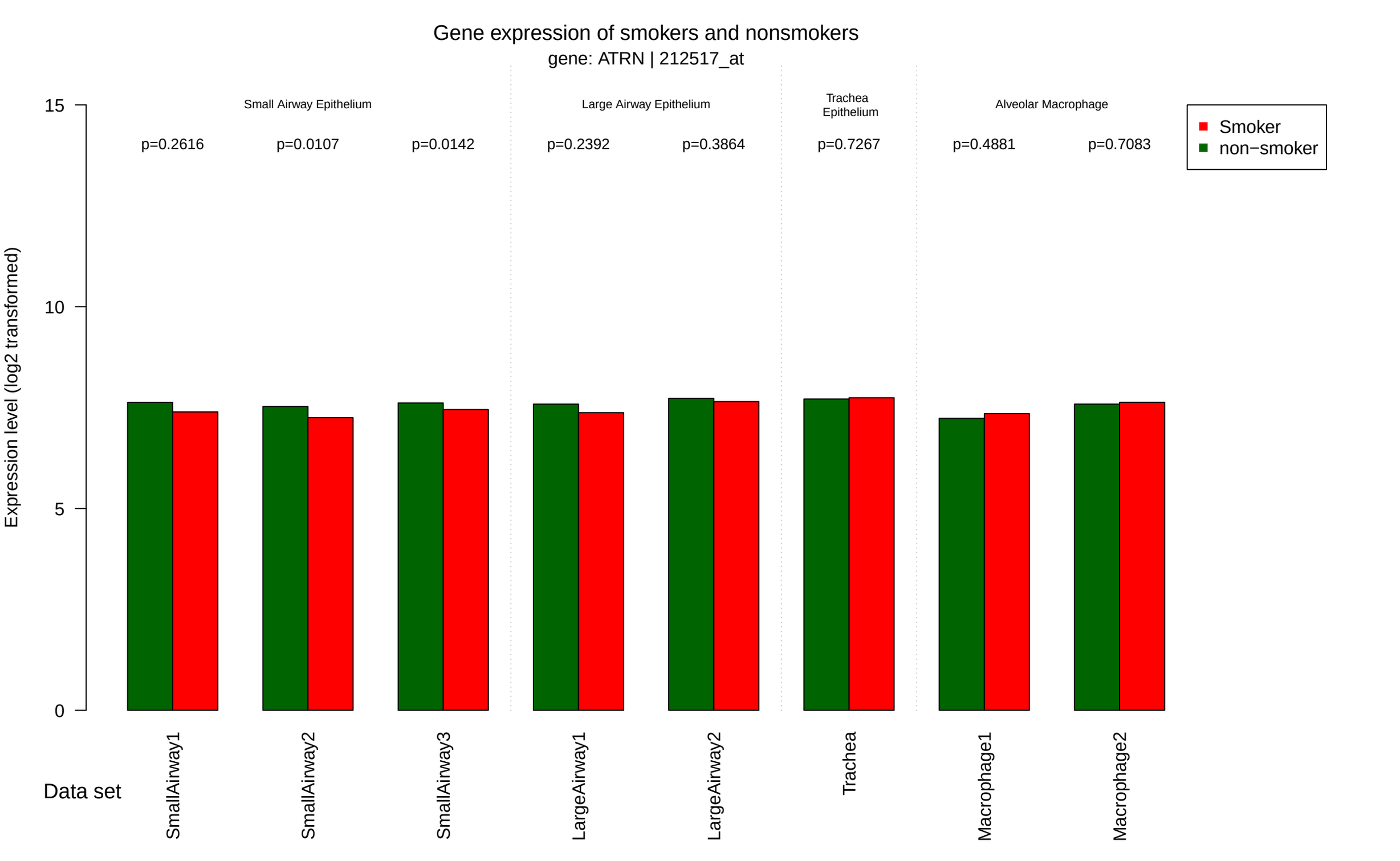
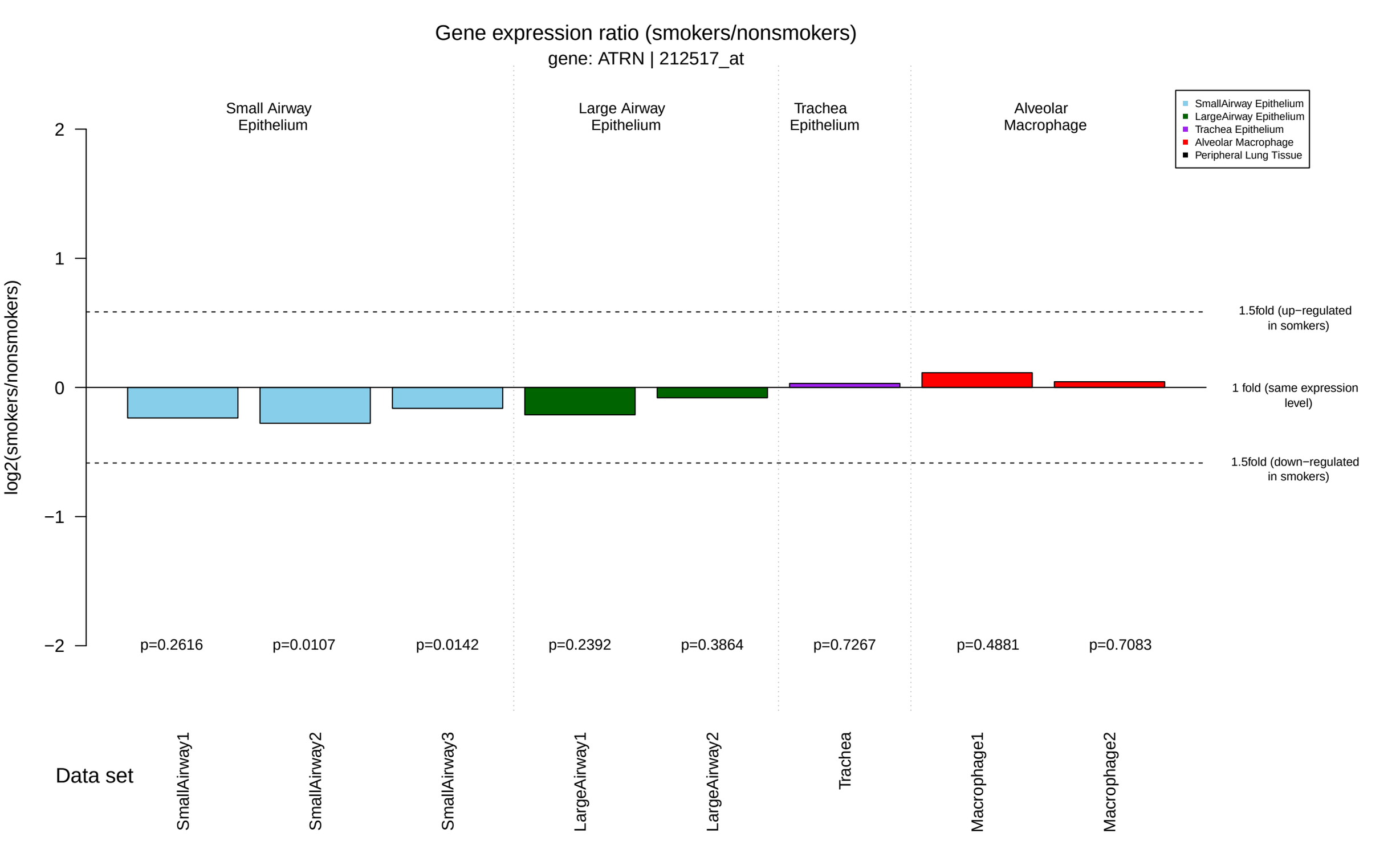
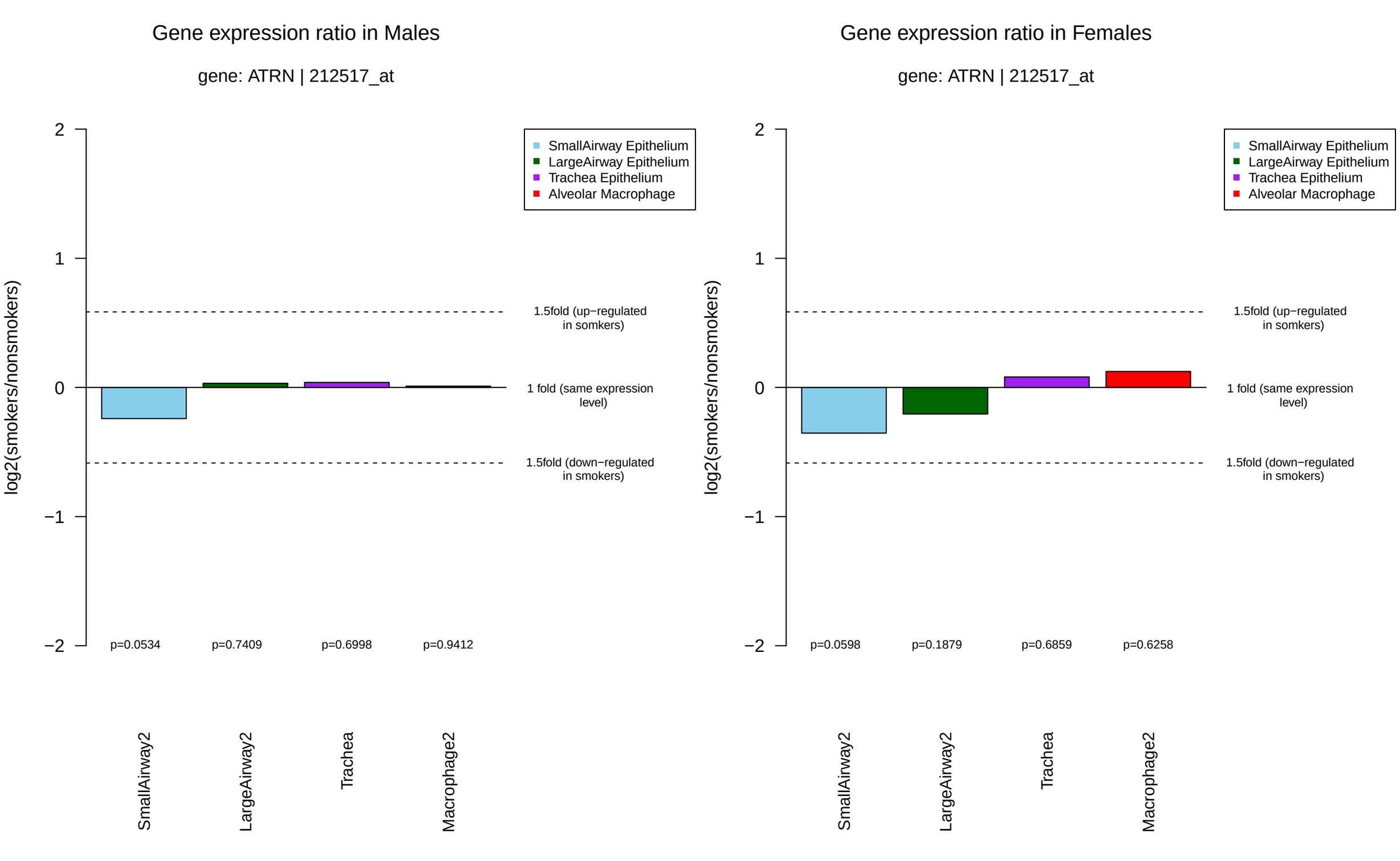
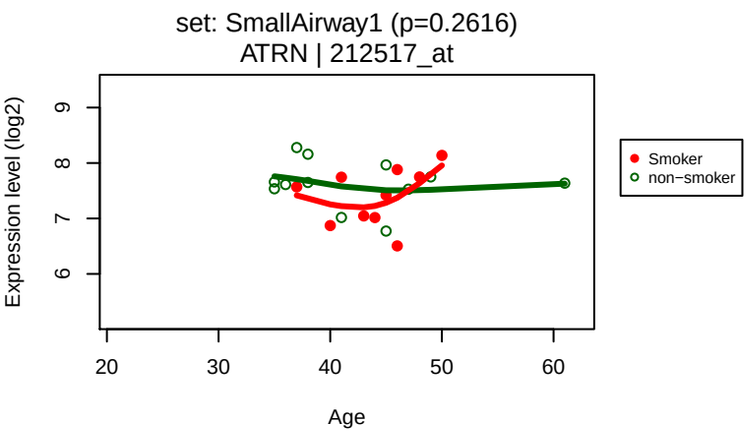
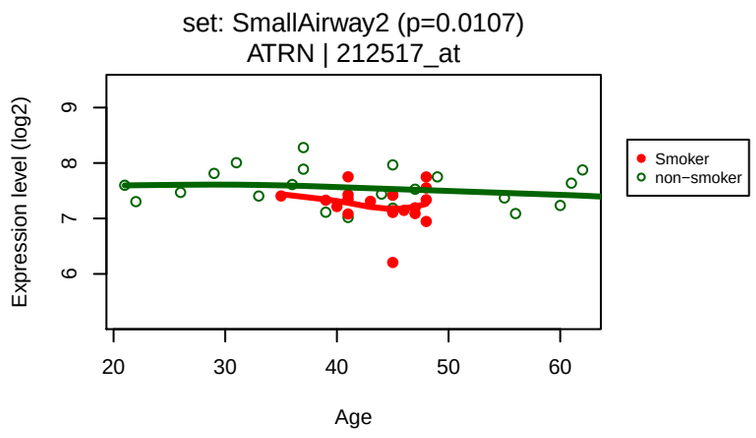
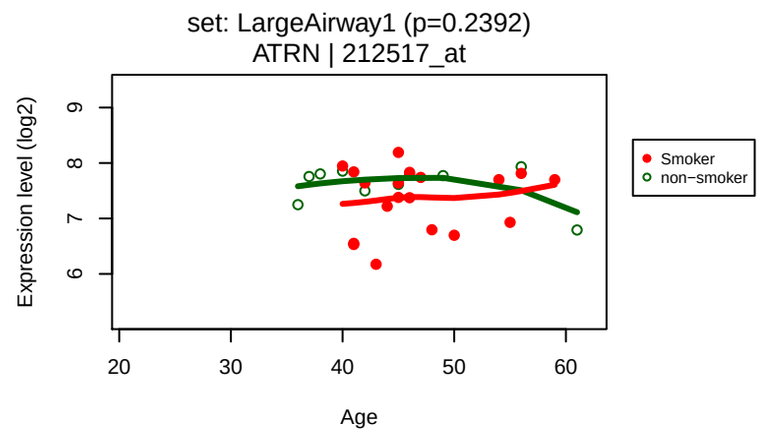

Gene expression of smokers and nonsmokers
gene: ATRN | 212517_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2616
p=0.0107
p=0.0142
p=0.2392
p=0.3864
p=0.7267
p=0.4881
p=0.7083
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

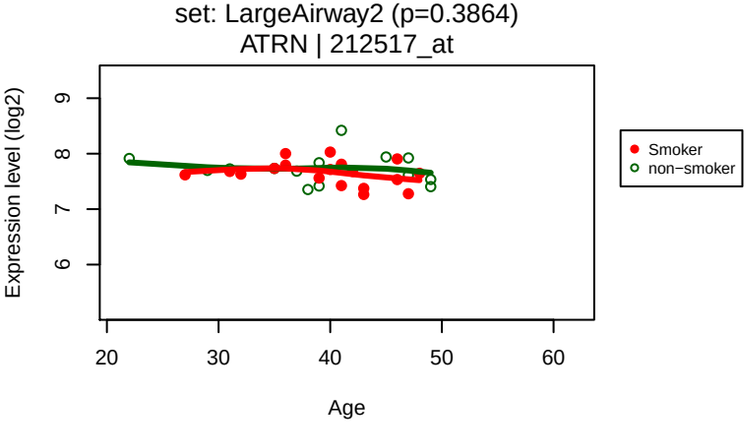

Gene expression ratio (smokers/nonsmokers)
gene: ATRN | 212517_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2616
p=0.0107
p=0.0142
p=0.2392
p=0.3864
p=0.7267
p=0.4881
p=0.7083
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

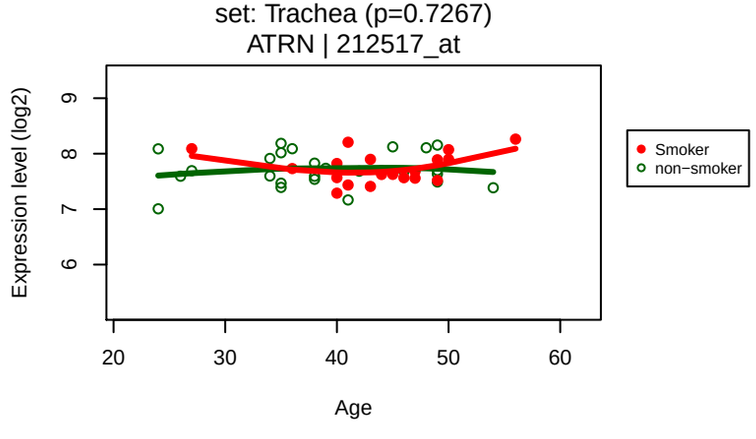

Gene expression ratio in Males
gene: ATRN | 212517_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0534
p=0.7409
p=0.6998
p=0.9412
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATRN | 212517_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0598
p=0.1879
p=0.6859
p=0.6258
SmallAirway2
LargeAirway2
Trachea
Macrophage2

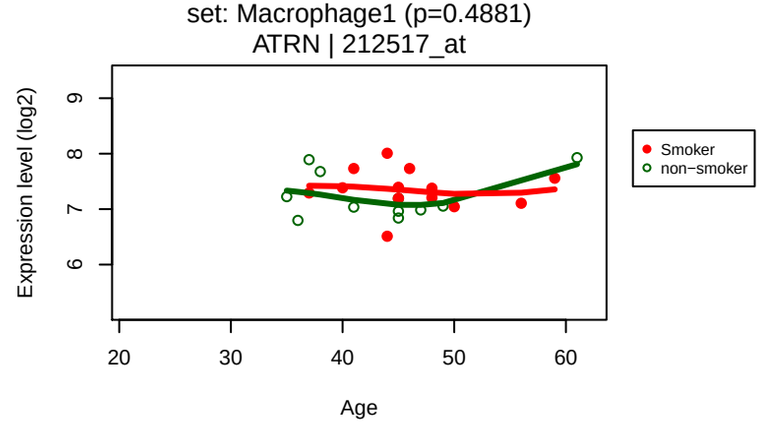

set: SmallAirway1 (p=0.2616)
ATRN | 212517_at
Expression level (log2)
Age
Smoker
non−smoker

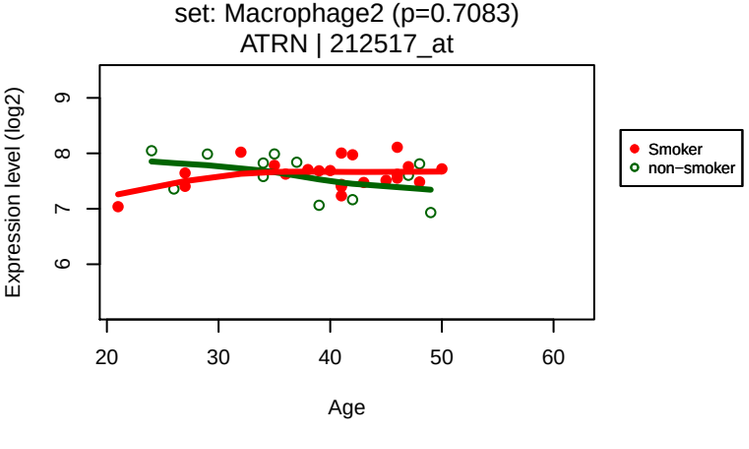

set: SmallAirway2 (p=0.0107)
ATRN | 212517_at
Expression level (log2)
Age
Smoker
non−smoker

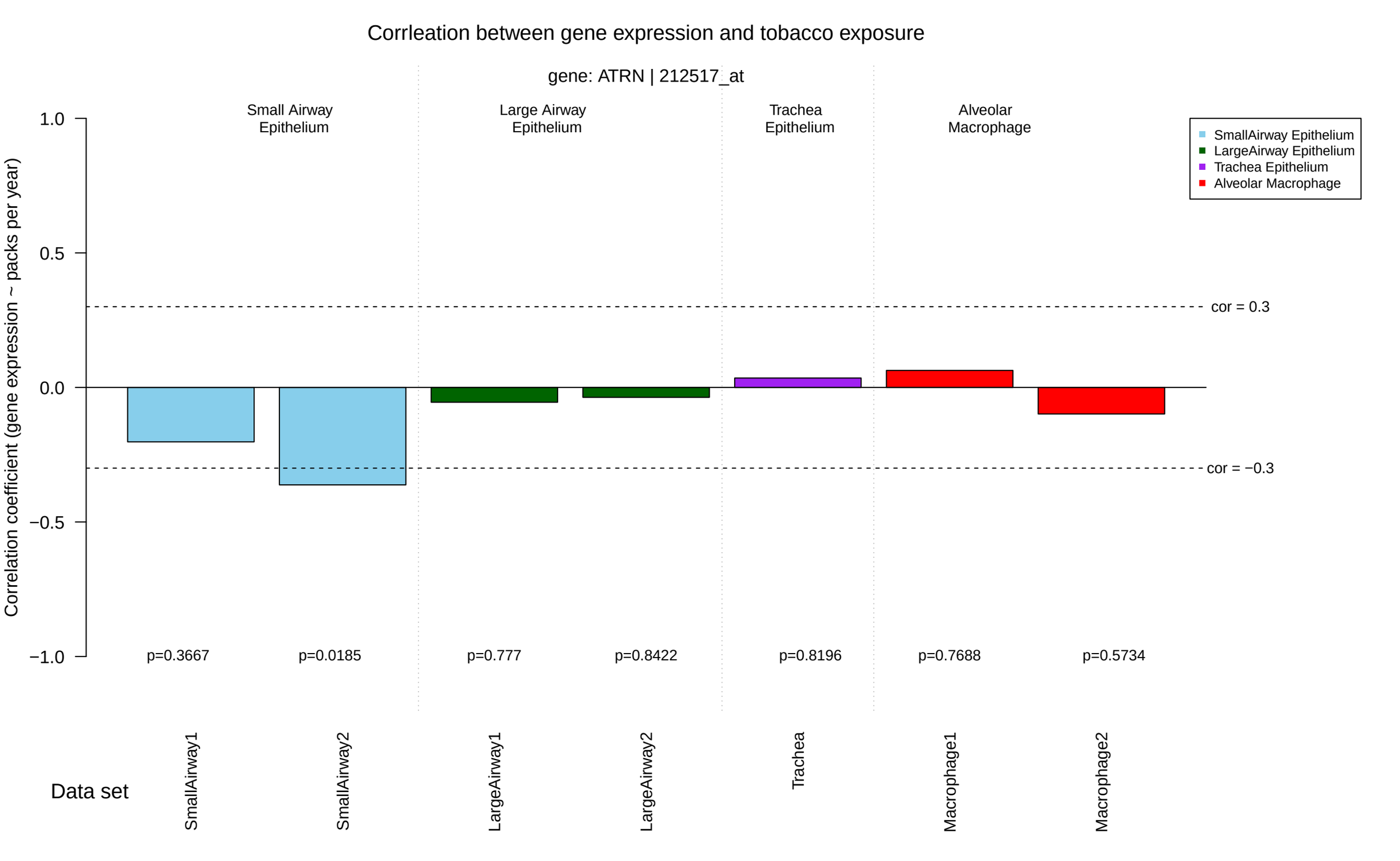

set: LargeAirway1 (p=0.2392)
ATRN | 212517_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3864)
ATRN | 212517_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7267)
ATRN | 212517_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4881)
ATRN | 212517_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7083)
ATRN | 212517_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATRN | 212517_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3667
p=0.0185
p=0.777
p=0.8422
p=0.8196
p=0.7688
p=0.5734
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2